1. What do you understand by Natural Language Processing?

Answer:

Natural Language Processing (NLP) is a branch of **Artificial Intelligence (AI)** that enables computers to understand, interpret, and generate human language. NLP combines linguistics, machine learning, and computer science to process text and speech data.

NLP is used in many real-world applications such as:

- Chatbots (e.g., ChatGPT)
- Language Translation
- Sentiment Analysis
- Voice Assistants
- Spam Detection
- Text Summarization

Example: When Google Translate converts English text into Hindi, it uses NLP techniques.

. What are the steps involved in solving an NLP problem?

Answer:

The main steps involved in solving an NLP problem are:

Step 1: Data Collection

Gather text data from sources such as websites, social media, documents, or databases.

Step 2: Text Preprocessing

Clean and prepare the text data by:

- Removing punctuation
- Converting text to lowercase
- Removing stop words
- Tokenization
- Stemming or Lemmatization

Step 3: Feature Extraction

Convert text into numerical form using techniques such as:

- Bag of Words (BoW)
- TF-IDF
- Word Embeddings

Step 4: Model Building

Train a machine learning or deep learning model on the processed data.

Step 5: Model Evaluation

Evaluate the model using metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score

Step 6: Deployment

Deploy the trained model for real-world use.

3. What is an Ensemble Method in NLP? With Example.

Answer:

An **Ensemble Method** is a technique that combines multiple machine learning models to improve prediction accuracy and overall performance.

Instead of relying on a single model, ensemble methods use several models and combine their predictions to produce better results.

Common Ensemble Methods:

1. Bagging
2. Boosting
3. Stacking

Example:

Suppose we want to classify movie reviews as **Positive** or **Negative**.

We train three models:

- Decision Tree
- Random Forest
- Logistic Regression

Predictions:

- Decision Tree → Positive
- Random Forest → Positive
- Logistic Regression → Negative

Using majority voting, the final prediction will be **Positive**.

This combination of multiple models is called an **Ensemble Method**.

Advantages:

- Higher accuracy
- Better generalization
- Reduced overfitting
- More robust predictions